



Risk factors for multiple metabolic syndrome components in obese and non-obese Japanese individuals

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ABSTRACT

Many studies have reported that even non-obese individuals have multiple metabolic syndrome (MetS) components, such as hypertension, hyperglycemia, and lipid abnormalities have a higher cardiovascular disease mortality rate and incidence. However, the risk factors for multiple MetS components in non-obese individuals have not been sufficiently clarified. This study compared risk factors, including overall lifestyle habits, for multiple MetS components possession between obese and non-obese individuals. A cross-sectional study was conducted using data from specific health checkups of 47,172 individuals (age, 40–64 years) who belonged to the health insurance societies of five manufacturing companies in Japan in 2015. Multiple logistic regression analysis was conducted in the non-obese and obese groups with multiple MetS components as the objective variable, and attributes, body weight change, and 11 lifestyle habits (smoking, exercise, diet, drinking, and sleep) as explanatory variables. For both groups, men, older age, current smoking, weight gain of ≥ 10 kg since age of 20, slow walking speed, fast eating speed, and greater amounts of alcohol consumption were risk factors for having multiple MetS components. The odds ratio of each risk factor, with the exception of walking speed and eating speed, tended to be higher in non-obese individuals than in obese individuals. The only risk factor specific to obese individuals was lack of regular exercise. These results suggest that almost all risk factors for possession of multiple MetS components were common to both obese and non-obese individuals, and the risk level of each risk factor tended to be higher in non-obese individuals.

1. Introduction

Metabolic syndrome (MetS), which is a complex of abdominal obesity, high blood pressure, dyslipidemia, and hyperglycemia, increases mortality and morbidity of cardiovascular diseases (CVD) and diabetes around the world (Noda et al., 2009; Saito et al., 2009; Mottillo et al., 2010). In Japan, MetS is defined as abdominal obesity plus two or more MetS components: hypertension, hyperglycemia, or lipid abnormalities (Metabolic Syndrome Diagnostic Criteria Committee, 2005). According to this viewpoint of visceral fat accumulation, the Ministry of Health, Labour and Welfare (MHLW) started conducting mandatory specific health checkups and specific health guidance for individuals aged 40–74 in communities and workplaces as a preventive measure against CVD and diabetes and lifestyle related diseases in 2008.

The specific health checkups consists of a medical examination and

self-administered questionnaire regarding lifestyle habits. Individuals whose waist circumference (male ≥ 85 cm, female ≥ 90 cm) or body mass index (BMI) ≥ 25 are defined as obese. Among those obese individuals, those who have two or more MetS components of hypertension, hyperglycemia, or lipid abnormalities, are provided active support for lifestyle habit improvement in specific health guidance. Thus, CVD and diabetes prevention measures are implemented in the obese people defined by both waist circumference and BMI, but not in the non-obese people in Japan.

On the other hand, it has been reported that not only obese, but also non-obese who have at least two MetS components of hypertension, hyperglycemia, or lipid abnormalities, had higher mortality and incidence of CVD and diabetes. In a community prospective cohort study, those with MetS components ≥ 2 , regardless of whether they were obese, had a higher hazard ratio for CVD mortality than those with 0 or 1

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components (Saito et al., 2009; Kadota et al., 2007). According to a community retrospective cohort study, individuals with MetS components ≥ 2 , regardless of whether they were obese, had a significantly higher hazard ratio for death from CVD than individuals with 0 or 1 components (Hamao et al., 2013). In another community and workplace prospective cohort study, individuals with more MetS components had higher hazard ratios for the development of stroke, regardless of whether they were obese (Ohashi et al., 2011).

Consequently, it is important to prevent the possession of multiple MetS components in both non-obese and obese individuals. However, only a few studies have investigated the risk factors for multiple MetS components in non-obese compared to obese individuals. In a cross-sectional study in Japan, physical inactivity was associated with the presence of multiple MetS components regardless of obesity status (Kaino et al., 2013). A prospective cohort study in the Netherlands also reported that physical inactivity and sedentary behavior were associated with the presence of multiple MetS components regardless of obesity status (de Rooij et al., 2016). In a cross-sectional study of Spanish workers, older, men, smokers, heavy drinkers, and those lacking physical exercise were linked to having multiple MetS components in both obese and non-obese individuals (Goday et al., 2016).

In this way, previous studies have implied that risk factors for having multiple MetS components might be similar in non-obese and obese individuals. However, in only few studies, overall lifestyle habits such as smoking, exercise, diet, drinking, and sleep have been analyzed. In addition, these studies have used only BMI to define obesity and not both BMI and waist circumference during specific health checkups in Japan. Thus, this study aimed to clarify the differences in risk factors including overall lifestyle habits for having multiple MetS components between obese and non-obese individuals, using data from specific health checkups in a large Japanese population.

2. Methods

2.1. Materials

We used data from specific health checkups of individuals belonging to five health insurance societies of manufacturing companies in 2015. Of the 97,518 individuals aged 40–64 who underwent routine annual specific health checkups based on the "standard program of the Specific Health Checkup and Guidance" (Ministry of Health, Labour and Welfare, 2013), 47,172 who had completed the following data from medical examination and self-administered questionnaires which were required to be answered during health checkups were analyzed. Anonymous data were provided by the Health Insurance Association, which approved its use for the current study. This study was approved by the Research Ethics Committee of the Faculty of Health and Sport Sciences of the University of Tsukuba, Japan (approval number: Tai 29–132; approval date: March 12, 2018).

2.2. Measures

Medical examination included attributes of sex, age, body mass index (BMI), abdominal circumference (around the navel), blood pressure (systolic and diastolic), blood sugar levels (fasting blood glucose [fasting of at least 10 hours required]), and lipid levels (triglycerides, high-density lipoprotein (HDL) cholesterol. Physical measurements, blood pressure measurements, and blood tests were performed on all individuals using the same techniques at registered health checkup institutions. The MHLW in Japan defines all inspection methods and stipulates quality control of all inspections under the "standard program of the Specific Health Checkup and Guidance" (Ministry of Health, Labour and Welfare, 2013). Thus, the accuracy of the test is guaranteed even if the operator performing the test is different.

Self-administered questionnaires, established in the "standard program of the Specific Health Checkup and Guidance" (Ministry of Health,

Labour and Welfare, 2013), include information on drugs (whether subjects were taking drugs for hypertension, diabetes, or dyslipidemia), body weight changes (weight gain of ≥ 10 kg since age of 20), and lifestyle habits (related to smoking, exercise, diet, drinking, and sleep). Body weight changes were assessed using a two-point "yes" or "no" rating scale pertaining to weight gain of ≥ 10 kg since age of 20 ("Have you gained over 10 kg from your weight since age of 20?"). Regarding lifestyle habits, current smoking ("Do you smoke cigarettes regularly?"), regular exercise ("Have you continued to engage in 30 minutes or more of a sweat-inducing exercise, two or more times per week, for one or more years?"), physical activity ("Do you walk or engage in physical activities with an equivalent intensity for one or more hours per day?"), walking speed ("Is your walking speed faster than someone with a similar age and same gender?"), eating late-evening meals (Do you eat meals 2 hours or less before bedtime, three or more times per week?), eating snacks after dinner ("Do you eat evening snacks other than your three regular meals, three or more times per week?"), skipping breakfast ("Do you skip breakfast three or more times per week?"), and having sufficient sleep ("Do you sleep well and enough?") were asked using a two-point "yes" or "no" rating scale. Eating speed ("Is your eating speed faster compared to others?") was rated on a three-point scale, with the following possible answers: "fast," "normal," or "slow." Frequency of drinking ("How often do you drink alcohol?") was also rated on a three-point scale as "every day," "sometimes," or "hardly ever/never." Amount of alcohol consumption ("How much amount of alcohol consumption per day on drinking days?") was rated on a four-point scale as "<180 mL/day", "180–360 mL/day", "360–540 mL/day", or " ≥ 540 mL/day" (A glass [180 mL] of refined sake (rice wine) is equivalent to a medium bottle [500 mL] of beer, 110 mL of shochu (alcohol content, 25%), a glass [double, 60 mL] of whiskey, and two glasses [240 mL] of wine).

Obesity, hypertension, hyperglycemia, and lipid abnormalities were defined according to the "standard program of the Specific Health Checkup and Guidance" (Ministry of Health, Labour and Welfare, 2013). Each criterion was determined as follows: obesity - waist circumference [male ≥ 85 cm, female ≥ 90 cm] and/or BMI ≥ 25 kg / m²; hypertension - systolic blood pressure ≥ 130 mmHg and/or diastolic blood pressure ≥ 85 mmHg, or receiving drug treatment for hypertension; hyperglycemia - fasting blood glucose ≥ 110 mg/dl or receiving drug treatment for diabetes; lipid abnormalities - triglycerides ≥ 150 mg/dl and/or HDL cholesterol < 40 mg/dl or receiving drug treatment for dyslipidemia (Japan Society for the Study of Obesity, 2016; Metabolic Syndrome Diagnostic Criteria Committee, 2005).

2.3. Statistical analysis

After observing the basic statistics of all variables in the non-obese and obese groups, the relationships between the number of MetS components of hypertension, hyperglycemia, and lipid abnormalities (" ≤ 1 " or " ≥ 2 "), and attributes, body weight changes, and lifestyle habits were analyzed using a Chi-square test. Multiple logistic regression analysis (forced entry method) was then conducted, with multiple MetS components (MetS components ≤ 1 : 0; MetS components ≥ 2 : 1) taken as the objective variable, and attributes, body weight change, and lifestyle habits as explanatory variables in each group. Reference explanatory variables (0) were taken to be "women" for sex, "40s" for age, and "no" for weight gain of ≥ 10 kg since age of 20. The following were used as references for good lifestyle habits: "no" for current smoking, "yes" for regular exercise, "yes" for physical activity, "fast" for walking speed, "slow" for eating speed, "no" for eating late-evening meals, "no" for eating snacks after dinner, "no" for skipping breakfast; "hardly ever/never" for frequency of drinking, "<180 mL/day" for amount of alcohol consumption, and "yes" for sufficient sleep.

IBM SPSS Statistics version 25 for Windows was used for data aggregation and statistical analyses. All reported values were considered statistically significant at a $p < 0.05$.

3. Results

3.1. Subject characteristics

The characteristics of the subjects (attributes, body weight change, and lifestyle habits) are shown in Table 1, wherein 28,720 were non-obese (60.9%) and 18,452 were obese (39.1%). Mean ages were 49.2 ± 6.2 years old in non-obese subjects and 50.1 ± 6.2 years old in obese subjects. Compared with non-obese subjects, a significantly higher percentage of obese subjects were men, in their 50s or 60s, had weight gained ≥ 10 kg since age of 20, were currently smoking, did not get regular exercise, were not physically active, had slow walking speeds, fast eating speeds, eating late-evening meals, skipping breakfast, drank alcohol every day, and drank large amounts of alcohol per day.

3.2. Relationship between possession of multiple MetS components and each explanatory variable

The relationships between possession of multiple MetS components with attributes, body weight change, and lifestyle habits are shown in Table 2. Both non-obese and obese individuals, compared to individuals with MetS components ≤ 1 , significantly higher percentages of those with MetS components ≥ 2 were men, in their 50s or 60s, had weight gained ≥ 10 kg since age of 20, had currently smoking, were fast eaters, did not eat snacks after dinner, had everyday drinking, and had greater amount of alcohol consumption. Among non-obese individuals only, significantly higher percentages had eating late-evening meals, skipping breakfast, and got sufficient sleep. Among obese individuals only, significantly higher percentages did not exercise regularly and had slow walking speed.

Table 1
Subject characteristics.

		All		Non-obese		Obese		p
		n = 47172		(n = 28720, 60.9%)		(n = 18452, 39.1%)		
		n (%) or Mean ± SD		n (%) or Mean ± SD		n (%) or Mean ± SD		
Attributes								
Sex	Men	36010	(76.3)	19336	(67.3)	16674	(90.4)	<0.001 ^a
	Women	11162	(23.7)	9384	(32.7)	1778	(9.6)	
Age		49.5 ± 6.2		49.2 ± 6.2		50.1 ± 6.2		<0.001 ^b
	40s	25078	(53.2)	15967	(55.6)	9111	(49.4)	<0.001 ^a
	50s	18673	(39.6)	10838	(37.7)	7835	(42.5)	
	60–64	3421	(7.2)	1915	(6.7)	1506	(8.1)	
Body weight change								
Weight gain of ≥10 kg since age of 20	Yes	19290	(40.9)	5714	(19.9)	13576	(73.6)	<0.001 ^a
	No	27882	(59.1)	23006	(80.1)	4876	(26.4)	
Lifestyle habits								
Current Smoking	Yes	12938	(27.4)	6985	(24.3)	5953	(32.3)	<0.001 ^a
	No	34234	(72.6)	21735	(75.7)	12499	(67.7)	
Regular exercise	Yes	11586	(24.6)	7354	(25.6)	4232	(22.9)	<0.001 ^a
	No	35586	(75.4)	21366	(74.4)	14220	(77.1)	
Physical activity	Yes	18246	(38.7)	11663	(40.6)	6583	(35.7)	<0.001 ^a
	No	28926	(61.3)	17057	(59.4)	11869	(64.3)	
Walking speed	Fast	23094	(49.0)	14787	(51.5)	8307	(45.0)	<0.001 ^a
	Slow	24078	(51.0)	13933	(48.5)	10145	(55.0)	
Eating speed	Slow	3644	(7.7)	2676	(9.3)	968	(5.2)	<0.001 ^a
	Normal	28480	(60.4)	18362	(64.0)	10118	(54.9)	
	Fast	15048	(31.9)	7682	(26.7)	7366	(39.9)	
Eating late-evening meals	Yes	16287	(34.5)	9132	(31.8)	7155	(38.8)	<0.001 ^a
	No	30885	(65.5)	19588	(68.2)	11297	(61.2)	
Eating snacks after dinner	Yes	6480	(13.7)	3895	(13.6)	2585	(14.0)	0.170 ^a
	No	40692	(86.3)	24825	(86.4)	15867	(86.0)	
Skipping breakfast	Yes	8416	(17.8)	4762	(16.6)	3654	(19.8)	<0.001 ^a
	No	38756	(82.2)	23958	(83.4)	14798	(80.2)	
Frequency of drinking	Hardly ever/never	8324	(17.7)	5151	(17.9)	3173	(17.2)	<0.001 ^a
	Sometimes	21387	(45.3)	13157	(45.8)	8230	(44.6)	
	Every day	17461	(37.0)	10412	(36.3)	7049	(38.2)	
Amount of alcohol consumption	<180 mL/day	21044	(44.6)	13976	(48.7)	7068	(38.3)	<0.001 ^a
	180–360 mL/day	15735	(33.4)	9406	(32.7)	6329	(34.3)	
	360–540 mL/day	7708	(16.3)	4058	(14.1)	3650	(19.8)	
	≥540 mL/day	2685	(5.7)	1280	(4.5)	1405	(7.6)	
Sufficient sleep	Yes	28026	(59.4)	17148	(59.7)	10878	(59.0)	0.104 ^a
	No	19146	(40.6)	11572	(40.3)	7574	(41.0)	
Cardiovascular risk factors								
MetS components	≤1	37505	(79.5)	25710	(89.5)	11795	(63.9)	<0.001 ^a
	≥2	9667	(20.5)	3010	(10.5)	6657	(36.1)	
BMI		23.4 ± 3.4		21.4 ± 1.9		26.5 ± 3.0		<0.001 ^b
Waist circumference		82.8 ± 9.2		77.2 ± 5.4		91.4 ± 6.9		<0.001 ^b
Systolic blood pressure		121.7 ± 15.5		118.4 ± 15.1		127.0 ± 14.6		<0.001 ^b
Diastolic blood pressure		76.9 ± 11.5		74.3 ± 11.1		81.1 ± 10.8		<0.001 ^b
Triglyceride		116.6 ± 95.7		96.5 ± 77.9		148.0 ± 111.2		<0.001 ^b
HDL-cholesterol		62.7 ± 16.5		67.4 ± 16.7		55.4 ± 13.3		<0.001 ^b
Fasting blood glucose		96.7 ± 16.4		93.8 ± 12.9		101.4 ± 20.0		<0.001 ^b

^a Chi-square test.

^b t-test.

Table 2
Relationships between possession of multiple MetS components and each explanatory variable.

		Non-obese (n = 28720)				Obese (n = 18452)					
		MetS components ≤ 1		MetS components ≥ 2		p	MetS components ≤ 1		MetS components ≥ 2		p
		(n = 25710, 89.5%)		(n = 3010, 10.5%)			(n = 11795, 63.9%)		(n = 6657, 36.1%)		
		n	%	n	%		n	%	n	%	
Attributes											
Sex	Men	16720	(65.0)	2616	(86.9)	<0.001	10415	(88.3)	6259	(94.0)	<0.001
	Women	8990	(35.0)	394	(13.1)		1380	(11.7)	398	(6.0)	
Age	40s	15091	(58.7)	876	(29.1)	<0.001	6533	(55.4)	2578	(38.7)	<0.001
	50s	9170	(35.7)	1668	(55.4)		4529	(38.4)	3306	(49.7)	
	60–64	1449	(5.6)	466	(15.5)		733	(6.2)	773	(11.6)	
Body weight change											
Weight gain of ≥10 kg since age of 20	Yes	4822	(18.8)	892	(29.6)	<0.001	8401	(71.2)	5175	(77.7)	<0.001
	No	20888	(81.2)	2118	(70.4)		3394	(28.8)	1482	(22.3)	
Lifestyle habits											
Current Smoking	Yes	5970	(23.2)	1015	(33.7)	<0.001	3622	(30.7)	2331	(35.0)	<0.001
	No	19740	(76.8)	1995	(66.3)		8173	(69.3)	4326	(65.0)	
Regular exercise	Yes	6554	(25.5)	800	(26.6)	0.200	2763	(23.4)	1469	(22.1)	0.036
	No	19156	(74.5)	2210	(73.4)		9032	(76.6)	5188	(77.9)	
Physical activity	Yes	10474	(40.7)	1189	(39.5)	0.196	4265	(36.2)	2318	(34.8)	0.071
	No	15236	(59.3)	1821	(60.5)		7530	(63.8)	4339	(65.2)	
Walking speed	Fast	13257	(51.6)	1530	(50.8)	0.452	5387	(45.7)	2920	(43.9)	0.018
	Slow	12453	(48.4)	1480	(49.2)		6408	(54.3)	3737	(56.1)	
Eating speed	Slow	2424	(9.4)	252	(8.4)	0.049	4630	(39.2)	2736	(41.1)	0.016
	Normal	16453	(64.0)	1909	(63.4)		6518	(55.3)	3600	(54.1)	
Eating late-evening meals	Fast	6833	(26.6)	849	(28.2)		647	(5.5)	321	(4.8)	
	Yes	8120	(31.6)	1012	(33.6)	0.024	4576	(38.8)	2579	(38.7)	0.950
Eating snacks after dinner	No	17590	(68.4)	1998	(66.4)		7219	(61.2)	4078	(61.3)	
	Yes	3610	(14.0)	285	(9.5)	<0.001	1745	(14.8)	840	(12.6)	<0.001
Skipping breakfast	No	22100	(86.0)	2725	(90.5)		10050	(85.2)	5817	(87.4)	
	Yes	4210	(16.4)	552	(18.3)	0.006	2323	(19.7)	1331	(20.0)	0.631
Frequency of drinking	No	21500	(83.6)	2458	(81.7)		9472	(80.3)	5326	(80.0)	
	Hardly ever/never	4754	(18.5)	397	(13.2)	<0.001	2054	(17.4)	1119	(16.8)	<0.001
	Sometimes	12002	(46.7)	1155	(38.4)		5427	(46.0)	2803	(42.1)	
Amount of alcohol consumption	Every day	8954	(34.8)	1458	(48.4)		4314	(36.6)	2735	(41.1)	
	<180 mL/day	12981	(50.5)	995	(33.0)	<0.001	4813	(40.8)	2255	(33.9)	<0.001
	180–360 mL/day	8311	(32.3)	1095	(36.4)		4018	(34.1)	2311	(34.7)	
	360–540 mL/day	3396	(13.2)	662	(22.0)		2148	(18.2)	1502	(22.6)	
Sufficient sleep	≥540 mL/day	1022	(4.0)	258	(8.6)		816	(6.9)	589	(8.8)	
	Yes	15283	(59.4)	1865	(62.0)	0.008	6916	(58.6)	3962	(59.5)	0.249
	No	10427	(40.6)	1145	(38.0)		4879	(41.4)	2695	(40.5)	

Chi-square test.

3.3. Factors related to possession of multiple MetS components

The results of the multiple logistic regression analysis are shown in Table 3. As a result of controlling mutual influences among explanatory variables, in both the non-obese and obese subjects, possession of multiple MetS components had a positive relationship with sex(men), age (older), weight gain of ≥10 kg since age of 20, current smoking, slow walking speed, fast eating speed, and great amount of alcohol consumption, but had a negative relationship with eating snacks after dinner and frequency of drinking. Only in obese individuals, possession of multiple MetS components was positively related to lack of regular exercise.

Higher odds ratios (ORs) of possessing multiple MetS components were observed for age(older), sex(men), weight gain of ≥10 kg since the age of 20, and amount of alcohol consumption in non-obese compared to obese individuals as follows: 50s (non-obese OR 3.09, obese OR 1.88) and 60–64 (non-obese OR 5.93, obese OR 2.81); men (non-obese OR 2.67, obese OR 1.91); weight gain of ≥10 kg since the age of 20 (non-obese OR 1.67, obese OR 1.45); current smoking of "yes" (non-obese OR 1.24, obese OR 1.16); amount of alcohol consumption of "180–360 mL/day" (non-obese OR 1.35, obese OR 1.21), "360–540mL/day" (non-obese OR 1.81, obese OR 1.44), and "≥540mL/day" (non-obese OR 2.50, obese OR 1.52)."

4. Discussion

To date, the risk factors for possessing multiple MetS components in non-obese individuals compared to obese individuals has not been sufficiently investigated. This study aimed to clarify differences in risk factors including overall lifestyle habits, for possession of multiple MetS components between obese and non-obese individuals, using data from specific health checkups done on a large Japanese population. Among the subjects in this study, 39.1% were obese and 60.9% were non-obese. This is similar to the results of the National Health and Nutrition Survey (obese with the same definition as this study: 42.1%, non-obese: 57.9%) (Ministry of Health, Labour and Welfare, 2017) and a large-scale workplace survey (obese with the same definition as this study: 37.1%, non-obese: 62.9%) (Federation of Health Insurance Unions, 2016). Thus, the distribution of obesity in the subjects of this study may be considered typical in Japan.

First, as a result of adjusting mutual influences among explanatory variables in multiple logistic regression analysis, men, older, weight gain of ≥10 kg since age of 20, current smoking, slow walking speed, fast eating speed, and greater amounts of alcohol consumption were risk factors independently related to possession of multiple MetS components in both non-obese and obese individuals. These results were consistent with a previous study reporting that men, older age, smoking, and heavy drinking were associated with multiple MetS components in

Table 3

Factors associated with possession of multiple MetS components (multiple logistic regression analysis).

		Non-obese			Obese		
		OR	95%CI	p	OR	95%CI	p
Attributes							
Sex	Women	1.00			1.00		
	Men	2.67	2.38–3.01	<0.001	1.91	1.69–2.16	<0.001
Age	40s	1.00			1.00		
	50s	3.09	2.83–3.38	<0.001	1.88	1.76–2.00	<0.001
	60–64	5.93	5.20–6.77	<0.001	2.81	2.51–3.15	<0.001
Body weight change							
Weight gain of ≥ 10 kg since age of 20	No	1.00			1.00		
	Yes	1.67	1.53–1.83	<0.001	1.45	1.35–1.56	<0.001
Lifestyle habits							
Current Smoking	No	1.00			1.00		
	Yes	1.24	1.14–1.36	<0.001	1.16	1.08–1.24	<0.001
Regular exercise	Yes	1.00			1.00		
	No	1.09	0.99–1.20	0.073	1.11	1.02–1.19	0.010
Physical activity	Yes	1.00			1.00		
	No	1.06	0.97–1.15	0.172	1.05	0.98–1.12	0.140
Walking speed	Fast	1.00			1.00		
	Slow	1.11	1.02–1.20	0.012	1.12	1.05–1.19	<0.001
	Slow	1.00			1.00		
Eating speed	Normal	1.16	1.00–1.34	0.047	1.13	0.97–1.30	0.108
	Fast	1.21	1.04–1.42	0.016	1.23	1.06–1.42	0.006
Eating late-evening meals	No	1.00			1.00		
	Yes	0.95	0.87–1.04	0.272	0.98	0.92–1.05	0.620
Eating snacks after dinner	No	1.00			1.00		
	Yes	0.79	0.69–0.91	<0.001	0.87	0.80–0.96	0.004
Skipping breakfast	No	1.00			1.00		
	Yes	1.06	0.96–1.19	0.255	1.04	0.96–1.12	0.380
Frequency of drinking	Hardly ever/never	1.00			1.00		
	Sometimes	0.84	0.74–0.96	0.012	0.79	0.71–0.87	<0.001
	Every day	1.02	0.88–1.17	0.818	0.82	0.74–0.91	<0.001
	<180 mL/day	1.00			1.00		
Amount of alcohol consumption	180–360 mL/day	1.35	1.22–1.50	<0.001	1.21	1.11–1.32	<0.001
	360–540 mL/day	1.81	1.61–2.05	<0.001	1.44	1.31–1.58	<0.001
	≥ 540 mL/day	2.50	2.12–2.96	<0.001	1.52	1.34–1.73	<0.001
Sufficient sleep	Yes	1.00			1.00		
	No	0.93	0.85–1.01	0.078	0.99	0.92–1.05	0.647
		Cox-Snell $R^2 = 0.071$			Cox-Snell $R^2 = 0.050$		
		Nagelkerke $R^2 = 0.146$			Nagelkerke $R^2 = 0.069$		

Multiple logistic regression analysis (forced entry method) was conducted with multiple MetS components (MetS components ≤ 1 : 0; MetS components ≥ 2 : 1) taken as the objective variable and attributes, body weight change, and lifestyle habits as explanatory variables in each group.

both obese and non-obese individuals (Goday et al., 2016).

On the other hand, the results of the study showed that lack of regular exercise was related to possession of multiple MetS components in only obese individuals, and physical activity had no relation to multiple MetS components possession in both obese and non-obese individuals. These results are not consistent with previous studies reporting that lack of regular physical exercise (Goday et al., 2016) and physical activity (Kaino et al., 2013; de Rooij et al., 2016) were associated with multiple MetS components in both obese and non-obese individuals. It might be because of the different measurements used for variables in this study compared to previous studies, which were defined in this study as follows: 'regular exercise' defined by 30 min or more of sweat-inducing exercise, two or more times per week, for one or more years; and 'physical activity' defined as walking or engaging in physical activity with an equivalent intensity for one or more hours per day. On the other hand, in previous studies, 'regular exercise' was defined by ≥ 2 h/week of physical exercise (Goday et al., 2016), and 'physical activity' was assessed from total energy and activity-specific expenditure (Kaino et al., 2013) or measured objectively by monitor devices (de Rooij et al., 2016). Thus, further verification of the relationship between exercise or physical activity and possession of multiple MetS components in non-obese compared to obese individuals is needed in future studies.

Second, the results of this study showed that ORs of older, men, greater amount of alcohol consumption, current smoking, and weight gain of ≥ 10 kg since age of 20, were especially higher in non-obese than

in obese individuals. Although previous studies have not clarified the risk level of each risk factor in non-obese individuals compared to obese individuals, the results of this study suggest that the risk level of having multiple MetS components due to these factors is higher in non-obese individuals than in obese individuals.

Non-obese men who were 50 and older had an especially higher risk of having multiple MetS components than obese individuals. The OR (versus the 40s) was 3.09 for those in their 50s and 5.93 for those in their 60s who were non-obese, whereas it was 1.88 for those in their 50s and 2.81 for those in their 60s who were obese. The OR of men versus women was 2.67 in the non-obese subjects and 1.91 in the obese subjects. Thus, measures to prevent the development of multiple MetS components seems to be necessary for non-obese men aged 50 and older.

Next, the OR (versus alcohol consumption of less than 180 mL/day) was 1.81 for 360–540 mL/day and 2.50 for ≥ 540 mL/day in non-obese individuals, whereas it was 1.44 for 360–540 mL/day and 1.52 for ≥ 540 mL/day in obese individuals. Current smoking status was also associated with a higher risk in non-obese individuals (OR 1.24) than in obese individuals (OR 1.16). Support measures with respect to addressing heavy alcohol consumption per day and current smoking are thus particularly necessary for non-obese individuals.

In addition, the risk of weight gain of ≥ 10 kg since age of 20 years was higher in non-obese (OR 1.67) than in obese individuals (OR 1.45). It has previously been reported that weight gain of ≥ 10 kg from early adulthood is associated with an elevated incidence of type 2 diabetes,

hypertension, and cardiovascular disease (Yan et al., 2017), weight gain from the age of 20 and increased BMI are associated with the development of diabetes (Kaneto et al., 2013), and weight gain of ≥ 10 kg over 20 years in individuals who were not obese in their 20s increases the risk for hypertension and diabetes in individuals in their 40s (Hatanaka et al., 2012). The findings of this study also demonstrated that weight gain of ≥ 10 kg from the age of 20 is linked to high risk for multiple MetS components possession in individuals who are currently not obese. Preventing long-term weight gain in young individuals with thin body types is thus important.

Third, according to the current study's results, eating snacks after dinner and occasional drinking decreased the risk for possessing multiple MetS components in both non-obese and obese individuals. Eating snacks after dinner, however, increases total daily energy intake, and late-night eating is known to lead to long-term weight gain in non-obese individuals (Suzuki et al., 2013). Thus, further investigation regarding the risk associated with eating snacks after dinner, including food content, energy intake, and time of eating, is needed. Previous studies have also reported that individuals who drink alcohol 3–4 times a week have a lower risk of developing CVD than individuals who drink less than once a week (Kenneth et al., 2003), and individuals who “sometimes/never drink” have a lower risk of possessing MetS components than individuals who “drinking every day” (Ohashi et al., 2011; Inoue et al., 2011). The present findings also suggest the possibility that occasional drinking is effective in lowering risk for MetS, but further studies combining frequency and amount of alcohol consumption are needed.

This study had the following limitations. First, as a cross-sectional study, this investigation could not clarify causal relationships; therefore, verification with a longitudinal study will be needed in the future. Second, the study subjects were individuals who belonged to company health insurance societies; thus, generalizing the study's results to the entire Japanese population is difficult. Third, the items analyzed in this study were limited to specific health checkup data; thus, verification using a wider range of objective data and survey data is also needed.

Despite the above limitations, this is the first study to compare risk factors, including overall lifestyle habits, for the possession of multiple MetS components between obese and non-obese individuals in a large Japanese population. The key strength of this study is that it elucidated specific risk factors for multiple MetS components in non-obese individuals and showed that improvements in these factors are needed for both obese and non-obese individuals.

5. Conclusions

This study showed that men, older age, weight gain of ≥ 10 kg since age of 20, current smoking, slow walking speed, fast eating speed, and greater amounts of alcohol consumption were risk factors for possession of multiple MetS components common to both obese and non-obese individuals. The risk level of each factor, with the exceptions of walking speed and eating speed, tended to be higher in non-obese individuals than in obese individuals. The only risk factor specific to obese individuals was lack of regular exercise.

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Declaration of competing interest

The co-author, Sakiko Ozawa is a health care consultant (Senior Researcher) at the Tokio Marine & Nichido Risk Consulting Co., Ltd. The other co-authors have no conflicts of interest to disclose.

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